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Controlled flight of high-thrust ultralight ion-propelled microrobot with integrated sensing

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Abstract

Ion wind technology holds potential for the miniaturization of flying microrobots, generating thrust without mechanical moving parts, enabling simplified and light designs free from high-frequency actuators. Nevertheless, two fundamental challenges have limited its practical implementation:

insufficient payload capacity and inadequate controllability, particularly in multi-degrees-of-freedom flight. Here, we demonstrated an inertial measurement unit-based closed-loop controlled flight of a light (36.7 mg) ion-propelled microrobot, achieving a thrust-to-weight ratio of 5:1 and 1-hour long-endurance tethered hovering without mechanical actuators. Experimental validation indicates that the control strategy enhances stability, with a decrease of 83.11% and 89.21% in the root mean square error of pitch and roll angle, respectively. Leveraging origami-inspired design and cost-effective metal-polymer composites, our manufacturing approach enables the rapid assembly of complex microrobots at a disposable cost, overcoming a key barrier to practical swarm deployment. The microrobot's high load capability allows it to carry a high-fidelity image sensor and a fiber Bragg grating sensor while maintaining sufficient maneuverability to complete predefined tasks such as environmental surveillance and material identification. This work introduces an approach to autonomous microrobotic swarm flight, suggesting potential applications in confined-space surveillance, disaster rescue, and hazardous environment exploration.

Introduction

In the challenging landscape of narrow, restricted access, or uncertain environments, flying microrobots provide advantages for exploration, rescue, inspection, and a spectrum of other critical tasks¹⁻³. These microrobots may access previously inaccessible areas due to their aerial maneuverability. Driven by progress in materials and manufacturing, the field of microrobots has witnessed significant attention and ongoing breakthroughs in the research and development of flying microrobots⁴⁻⁷. Based on their propulsion mechanisms, flying microrobots are generally classified into three main types: propeller-driven fixed-wing^{8,9}, rotorcraft¹⁰⁻¹³, and flapping-wing¹⁴⁻

¹⁷. Unlike conventional propulsion systems, ion-propelled robots leverage ion wind for flight, representing an addition to the field of flying microrobots^{18,19}. Ion wind propulsion has been confirmed for use in externally powered microrobots¹⁸, and untethered flight with ion wind has also been achieved in larger aircraft equipped with fully integrated onboard power systems¹⁹. Ion wind itself, as a natural propulsion mechanism, provides a streamlined and biomimetic solution for microrobot design. The straightforward generation of precisely controlled airflow via ion wind technology facilitates the circumvention of traditional propulsion limitations, introducing emerging pathways for creating lightweight, maneuverable, versatile systems^{20,21}.

Corona discharge is the primary mechanism for ion-propelled flying microrobots, where strong electric fields ionize gas molecules, generating ion wind that subsequently produces thrust²². The use of ion wind in flying microrobots provides distinct advantages, as it negates the requirement for precision-engineered rotating components or flapping wings, thereby reducing system complexity, size, and weight²³⁻²⁵. This simplification enhances reliability and durability while also considerably reducing noise levels due to the absence of high-frequency motion. Ion-propelled flying microrobots are commonly centimeter-scale devices with masses of 30-50 mg and a thrust-to-weight ratio of around 3²⁶. Due to their limited payload capacity, these robots struggle to carry even lightweight payloads such as Microelectromechanical Systems (MEMS) devices, which restricts their practical applications. Enhancing the thrust-to-weight ratio has emerged as a main challenge for ion-propelled flying microrobots. Addressing this challenge demands a two-pronged optimization strategy: enhancing thrust while simultaneously reducing mass. On the one hand, thrust enhancement is determined by factors such as electrode configuration²⁷⁻²⁹, electrode spacing, and applied voltage³⁰⁻³². Although experiments have confirmed that thrust increases with voltage, further increases in voltage face technical and safety barriers. Therefore, optimizing

structure configuration, such as needle-wire, needle-ring, and needle-mesh, represents a more viable approach for thrust improvement³³⁻³⁵. On the other hand, in the pursuit of mass reduction, the use of light material types, as opposed to conventional materials like SOI silicon layers^{26,36} or metal foil³⁷, holds the key to minimizing weight without compromising structural integrity. By selecting different materials and optimizing structural design, our recent study reported an ion wind thruster weighing 16 mg that achieved takeoff with a thrust-to-weight ratio of 5.5:1³⁸. Another persistent challenge for ion-propelled flying microrobots lies in achieving precise controllability. Three main challenges impede the controllability of this flying microrobot. The first challenge is the nonlinear relationship between voltage and thrust, which is further complicated by its susceptibility to disturbances such as temperature, humidity, and pressure. The second challenge stems from the microrobot's low inertia and high sensitivity. Its low inertia makes it highly sensitive to thrust fluctuations, where minor thrust variations can lead to serious attitude deviation. Manufacturing and assembly present the third challenge for ion-propelled flying microrobots, as current methods involve costly materials like silicon-on-insulator (SOI) wafers, high operational expenses for cleanroom facilities and advanced equipment, and time-consuming manual assembly of fragile microcomponents. The lack of simplified and standardized manufacturing and testing protocols amplifies process variations, leading to mismatches between designed and actual inertial properties in microrobots that ultimately compromise control stability. Current research strategies to address these challenges focus primarily on enhancing passive stability through structural design^{39,40} or developing intelligent algorithms for stabilization without actual flight validation⁴¹⁻⁴³. However, robust closed-loop control implementations of ion-propelled flying microrobots—particularly in real-world experimental validations—remain underdeveloped. Although we have experimentally verified the feasibility of using multiple distributed ion thrusters

for attitude control⁴⁴, to date, no fully controllable and stable systems with inertial measurement unit (IMU)-based closed-loop control have been demonstrated, thus hindering their practical deployment compared to other types of flying microrobots⁴⁵⁻⁴⁷.

This paper proposes a light, low-cost ion-propelled microrobot with a high thrust-to-weight ratio for IMU-based closed-loop controlled flight. For its construction, the developed flying microrobot utilizes metal-coated polyethylene terephthalate (PET) film, capitalizing on its combination of metal conductivity, polymer flexibility, light weight, and low cost to achieve further weight reduction, easy assembly, robustness, and cost-effectiveness. Through structural optimization design guided by multiphysics simulations, the microrobot achieves dimensions of 37.7 mm × 37.7 mm and a mass of 36.7 mg. Notably, the proposed flying microrobot features a low material cost (under \$0.10 USD) and rapid assembly (less than 30 minutes), integrating onboard sensors and a circuit board. With four optimized ion wind thrusters arranged in a cross pattern, the microrobot achieves a thrust-to-weight ratio of 5:1, enabling it to carry various lightweight payloads, such as the IMU sensor, the image sensor, and the fiber Bragg grating (FBG) sensor. Controlled flight is achieved via a closed-loop attitude-control method, with functionality validated through square-path and triangle-path trajectory tracking. Experimental results also demonstrate that the microrobot maintains high maneuverability when carrying an image sensor and an FBG sensor, enabling image recording and material detection tasks, respectively.

Results

Design and assembly of the flying microrobot

Fig. 1a provides an overview of the proposed flying microrobot, highlighting its compact design (37.7 mm × 37.7 mm in length/width, 36.7 mg total mass) alongside a scale comparison with a US one-cent coin. An exploded view showing detailed components of the microrobot is presented in

Fig. 1b. It consists of four groups of ion wind thrusters, an IMU sensor integrated on a flexible printed circuit board (PCB), and structural support frames. The flexible PCB-mounted onboard IMU weighs 30 mg and is positioned at the microrobot's geometric center to accurately monitor flight attitude parameters.

The microrobot achieves propulsion through electrohydrodynamic (EHD) ion wind generation, where asymmetric electrode configurations create macroscopic airflow via corona discharge. A unit of the ion wind thruster, located at one edge of the square configuration, comprises a group of emitters and collectors that generate thrust through ion wind. The working process of the thruster is illustrated in Fig. 1c. When a high voltage (HV) difference is applied between the emitter and collector electrode, it induces a corona discharge in the air molecules between them, generating a large number of excited ions near the sharp emitters. Under the influence of the electric field between the electrodes, these ions accelerate and move towards the collectors. In transit towards the collectors, the accelerated ions collide with surrounding neutral air molecules, transferring momentum and creating an ion wind that generates thrust.

As demonstrated by the fundamental principles of ion wind propulsion, the key advantage of ion-propelled microrobots stems from their inherently simplified architecture, which eliminates the need for complex mechanical components. Rotorcraft microrobots require heavy electromagnetic motors to control the rotors, making it challenging to reduce their overall weight to sub-gram levels. Although flapping-wing microrobots are amenable to miniaturization, they still rely on complex mechanical actuators and transmission mechanisms to drive high-frequency flapping motions, which may limit their thrust-to-weight ratios. In contrast, the ion-propelled flying microrobot developed in this work utilizes PET polymer films to achieve a thrust-to-weight ratio of 5:1, representing an improvement over existing flying microrobot platforms^{2,11,12,14-17,23-}

26,36,43,46,48-54 in this regard (Fig. 1d). Details of the robots are provided in Supplementary Table 8. Furthermore, the integration of an IMU enables the robot to achieve attitude control and subsequently realize trajectory tracking, surpassing the limitations of previous ion-propelled flying microrobots that were capable only of basic takeoff and landing, or solely theoretical control analysis³⁶. With its high thrust-to-weight ratio, the flying microrobot demonstrates sufficient payload capacity, enabling integration of diverse functional modules, including image sensors and FBG sensors. The ability to adapt to specific mission requirements makes this microrobot particularly suitable for applications spanning infrastructure inspection, terrain mapping to parcel delivery, disaster rescue in harsh environments, and even deep-space exploration missions. (Fig. 1e).

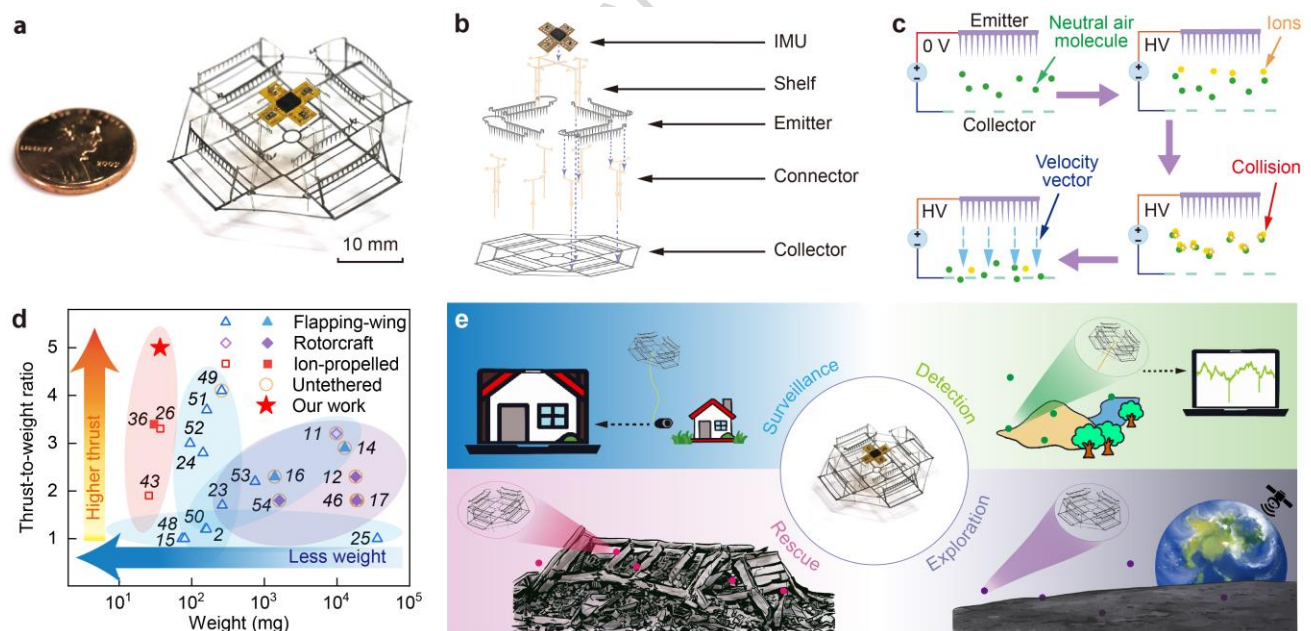


Fig. 1. Design, principle, and application of the controlled ion-propelled flying microrobot.

(a) Optical image of a 36.7-mg prototype alongside a US one-cent coin with a wingspan of 37.7 mm. (b) Schematic of the proposed flying microrobot's structure: emitters, collectors, structural connectors, central mounted shelf, inertial measurement unit (IMU) integrated on

flexible PCB. (c) Schematic illustrating the principle of the ion wind generation of the microrobot, with HV denoting high voltage. (d) Comparison of mass without power system and control electronics, thrust-to-weight ratio, and the presence of an IMU with different flying microrobots^{2,11,12,14-17,23-26,36,43,46,48-54}. The solid symbol indicates with an IMU, while the hollow symbol indicates without an IMU. Hollow circles indicate untethered flight. (e) A concept schematic illustrates the use of the ion-propelled flying microrobot in areas such as surveillance, detection, rescue, and exploration.

Characterization and analysis of ion thrusters' performance

A basic actuating unit for the developed microrobot, comprising two needle emitter structures and seven collector structures on each side of its square structure, is investigated. To describe the distribution of the ion wind, a multiphysics coupling computational method is employed, which couples the electrostatic field, laminar flow field, and rarefied mass transfer field. The three-dimensional (3D) computational model, as shown in Fig. 2a, is constructed using Comsol Multiphysics to quantitatively calculate the velocity field of the ion wind generated by the developed ion-propelled flying microrobot. A voltage difference applied across the emitter and collector electrodes induces the corona discharge, generating a substantial amount of ionized particles. The rarefied mass transfer field is introduced to calculate the motion of activated ions, while the flow field captures the macroscopic air movement. The ion-neutral molecule interaction is simulated by applying a Coulomb force-derived volume force to the air molecules. Fig. 2a demonstrates that the calculated airflow accelerates between the emitters and collectors, maintaining stable flow directionality, and further illustrates that varying the applied voltage (5 kV, 5.5 kV, and 6 kV) on the ion thrusters results in different airflow velocity profiles. The simulation results show that when the voltage difference between the emitter and collector rises to 6 kV, the

velocity of the ion wind reaches the peak (1.71 m/s). The 3D simulation results further indicate that applying varying voltage differences to individual thrusters generated an uneven wind speed distribution, leading to a variation in thrust. This thrust variation can effectively be converted into a control torque, providing a theoretical basis for the attitude control of the flying microrobot. To experimentally validate the simulation results, we tested the ion-propelled flying microrobot in a wind tunnel (Fig. 2b and Supplementary Movie 1). The smoke stream can pass through the microrobot when the ion wind is off. The ion wind generation was controllably activated and terminated by modulating the voltage difference between the emitters and collectors. A distinct impact of the ion wind on the lateral airflow confirmed the generation of thrust: smoke, initially flowing horizontally when corona discharge created the ion wind, was deflected downward.

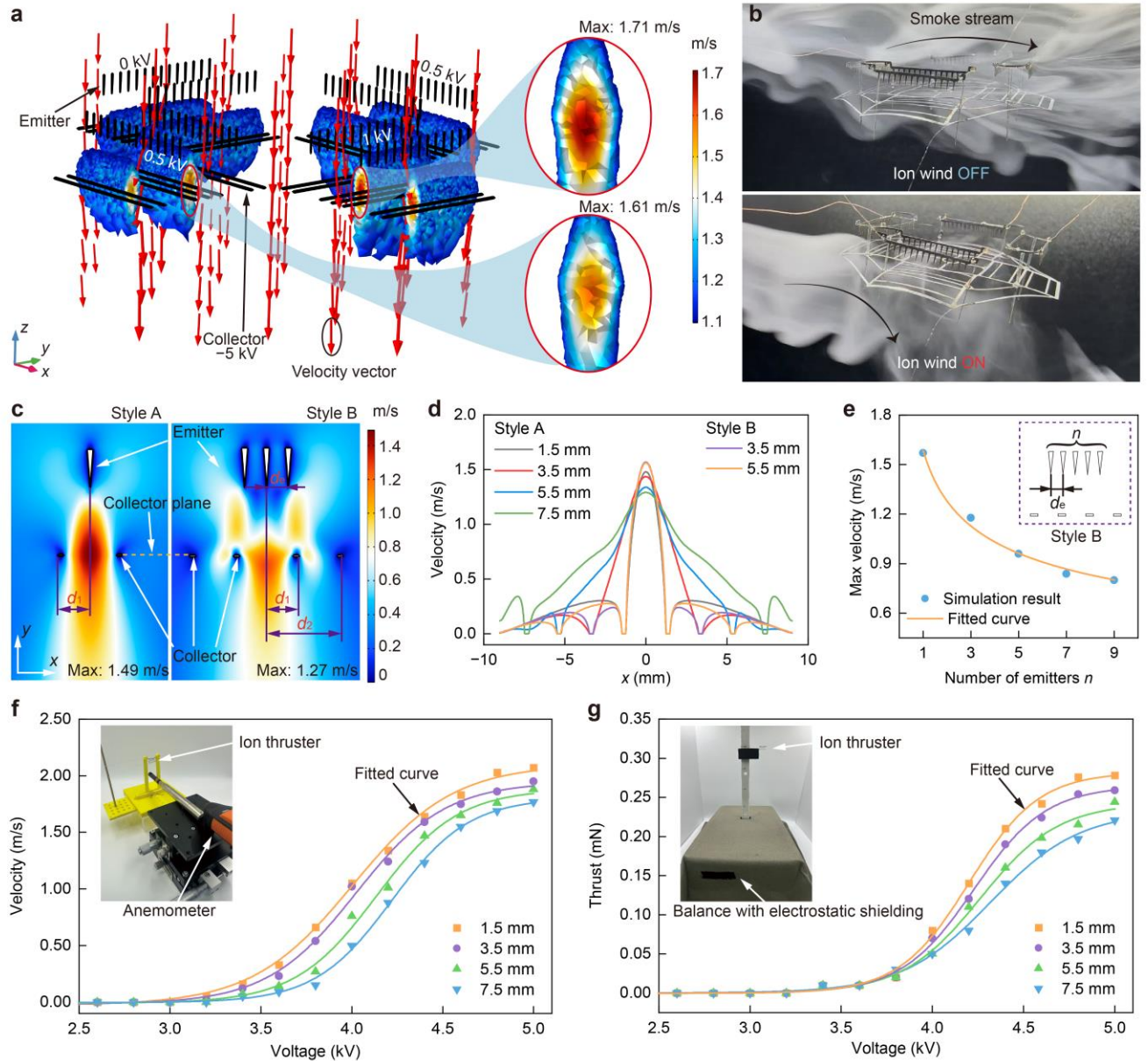


Fig. 2. Simulation and experimental validation of the performance of the designed thruster.

(a) Calculated 3D ion wind velocity field of the ion-propelled flying microrobot with varying applied voltages (5 kV, 5.5 kV, and 6 kV) across different thrusters. (b) Wind tunnel test demonstrating the visualization of the airflow in the absence and presence of ion wind generated by the microrobot. (c) Calculated 2D velocity field of ion wind generated by two configurations of the collector plane (style A and B), where $d_1 = 1.5$ mm, $d_2 = 3.5$ mm, $d_e = 1$ mm, and the voltage

difference is 5.5 kV. **(d)** Velocity distribution of ion wind showing the influence of different design parameters, demonstrating that the best structure design is $d_1 = 1.5$ mm and $d_2 = 3.5$ mm (style B). **(e)** Relationship between maximum wind velocity and number of emitters for style B, indicating no increase in maximum central velocity with additional emitters. **(f)** Experimental characterization of the wind velocity versus applied voltage for varying distances d_1 . Inset: anemometer for the wind velocity measurement. **(g)** Experimental characterization of the thrust versus applied voltage for varying distances d_1 . Inset: balance with electrostatic shielding to measure the lifting force.

As the primary figure of merit for flying microrobot performance, the thrust-to-weight ratio can be maximized through coupled optimization of emitter geometry and collector architecture. The optimization to improve the velocity of ion wind, increase thrust, and reduce structural mass is conducted through both simulations and experiments. Although increasing the distance d_g is an effective way to enhance lift, we chose not to increase the structure's height due to the negative impact on flight stability. A taller structure adversely affects stability by elongating the frame and raising the center of gravity. Then the simulation-based optimization employs two-dimensional (2D) models evaluating two alternative configurations (Fig. 2c). Both models feature needle-shaped emitters and rectangular cross-section collectors. The distance d_g is set to 4 mm. While style A employs a single emitter with two symmetric collectors, style B adds two outer collectors beyond the original pair. Fig. 2d shows the velocity distribution corresponding to different spacings d_1 between the emitter and the nearest collectors under a 5.5 kV voltage difference. Results show that the maximum wind speed on the collector plane occurred on the centerline of the emitters-collectors unit, both for style A and style B, with a single emitter. Simulation results indicate that when the spacing of collectors is 1.5 mm, the ion wind velocity on the collector plane

reaches 1.49 m/s. As the spacing increases, the maximum ion wind velocity on the collector plane decreases. Further investigation reveals that increasing the number of collectors alters the wind speed distribution. As shown in the wind speed profile for style B in Fig. 2d, when extra collectors are added to both sides of a pair of collectors, the ion wind flow field shows a distinct convergence, resulting in a higher wind speed, which reaches 1.57 m/s at maximum. By comprehensively considering the enhancement of wind speed and the reduction in mass, the combination of $d_1 = 1.5$ mm and $d_2 = 3.5$ mm is determined as the optimal design solution. Furthermore, the simulation of the emitters' numbers shows that the addition of extra emitters does not result in an increase in the maximum central wind speed with the same design of the collector structure, as illustrated in Fig. 2e. The explanation is as follows. When a single emitter is arranged symmetrically along the axis, the system can effectively absorb the ions around and generate a stable corona discharge. However, when multiple emitters are arranged symmetrically, the number of ionized air molecules is nearly the same as in the case of a single emitter. This observation can be attributed to the fact that when a single emitter is aligned along the thruster's central axis, it effectively achieves maximum local ionization density by activating nearly all available air molecules within its effective range, while maintaining stable corona discharge. However, multiple emitter arrangements demonstrate no net gain in total ionized molecules compared to the single-emitter case. Each ion wind thruster demonstrates well-defined critical corona discharge voltages, which determine the onset of stable ionization and thrust generation. Upon reaching the critical voltage, the population of ionizable molecules approaches saturation. Multiple emitters not only fail to increase ion production beyond this saturation limit but also degrade the thrust-to-weight ratio due to an increase in mass. Consequently, a single emitter configuration delivers optimal system efficiency.

The final configuration uses style B with a single row of emitter ($d_1 = 1.5$ mm, $d_2 = 3.5$ mm) as one unit. To enhance thrust, the microrobot's thruster incorporates two emitter-collector units arranged side by side. As shown in Fig. 2a, four thrusters form the ion-propelled microrobot. Experimental testing was conducted to determine the impacts of the spacing between the collectors on the ion wind velocity and thrust under actual working conditions. An anemometer was used to measure the ion wind velocity generated by the ion wind thruster. Furthermore, a precision balance was also used to measure the lifting force produced by the ion wind thrust, indicated by a reduction in weight. The thruster, mounted on a raised platform connected to a precision balance (resolution: 0.1 mg), was isolated within a Faraday cage-like enclosure. This electrostatic shielding prevented ionic wind interference and stray charges from compromising the balance's internal electronics and measurement accuracy. Varying the distance d_1 between the two inner collectors allowed us to obtain ion wind velocity and thrust for various thruster designs. According to the experimental data presented in Fig. 2f and g, a distance d_1 of 1.5 mm yielded the highest measured ion wind velocity (2.0 m/s) and thrust (0.28 mN) across the tested designs. The configurations of style B were also tested. The data are shown in Supplementary Fig. 13. The experimental results are in agreement with the simulation analysis that a compact structure with extra collectors yielded higher wind velocity and thrust. Based on the findings, the thruster's final structural parameters are set as $d_1 = 1.5$ mm and $d_2 = 3.5$ mm to achieve optimal performance. The optimal thruster generates a thrust of 0.31 mN, with a size of 12 mm $\times$ 12 mm $\times$ 4 mm, resulting in a thrust density of 538 N/m³, which is higher than that of the Airbus A320 (31.5 N/m³) and wire-wire ion wind thruster (110 N/m³)⁵⁵. It should be noted that this data was obtained under quasi-static loading conditions. Increasing the voltage ramp rate is beneficial for enhancing the thrust, and further details are provided in Supplementary Information.

Design of the configuration and characterization of materials

The thruster's modular design permits flexible arrangement into multiple configurations, allowing the construction of microrobots with customized flight characteristics. Representative microrobot configurations include triangular, quadrilateral, and pentagonal geometries (see Supplementary Fig. 2 for prototypes). To simplify the control algorithm and achieve decoupling of attitude control, this study employs a symmetrical cross-shaped arrangement of four thrusters, as illustrated in Fig. 3a. By adjusting the thrust of each ion thruster, the torque distribution generated by the ion wind on the fuselage of the microrobot can be altered, enabling active attitude control. This arrangement not only facilitated the balance of thrust and weight but also effectively improved the stability of the robot. Due to the symmetry of the structure, the thrust on each side can be simplified to an equivalent concentrated force of magnitude F_i , with the equivalent action point at O_{fi} . Given the negligible damping effects in practical conditions, the relationship between the magnitude of the ion wind thrust and the angular accelerations of pitch and roll ($\ddot{\theta}$ and $\ddot{\phi}$) under small angle conditions can be derived as follows:

$$\ddot{\theta} = \frac{(F_1 - F_3)l_f}{I} \quad (1)$$

$$\ddot{\phi} = \frac{(F_4 - F_2)l_f}{I} \quad (2)$$

where l_f is the arm length from the center of mass to the equivalent action point, and I is the rotational inertia of the structure about the x-axis and y-axis.

Fabrication material impacts the performance of flying microrobots, requiring optimized mechanical properties to achieve both flight capability and structural integrity. Ideal materials must combine high stiffness, high thrust-to-weight ratios, and minimal density to satisfy the demands of high reliability and thrust-to-weight ratio. Furthermore, high elasticity and flexural strength are

essential to resist thrust-induced bending moments while maintaining aerodynamic stability during aggressive maneuvers. Departing from conventional titanium or silicon-based microrobot fabrication, we propose a synergistic integration of metallic conductivity and polymeric advantages through metallized PET films. This hybrid material system combines the electrical functionality of thin metal coatings with PET's intrinsic high elasticity, high stiffness, ultralight weight, and low-cost processability, etc. The resulting microrobots achieve over 60% mass reduction compared to titanium counterparts while exhibiting better robust structural integrity, simplified assembly via laser-cuttable layers, and scalable manufacturing. Three emitter variants were fabricated using distinct materials: titanium (metallic), silicon (inorganic non-metallic), and metal-coated PET (hybrid polymer). Comparative destructive testing was subsequently conducted to evaluate their mechanical performance under loading conditions, with complete experimental results presented in Fig. 3b and Supplementary Movie 2. To compare their mechanical properties, three different emitter materials were tested by applying a force with a 5 g weight. The silicon emitter fractured immediately upon loading. While the metal film emitter did not fracture, it underwent substantial permanent plastic deformation. In contrast, the metallized PET film demonstrated better performance, maintaining its linear elasticity and recoverability even under large deformations, which confirms its high strength, robustness, and reliability. Therefore, the 50- μm -thick metallized PET film was utilized as the optimal fabrication material, offering a combination of ultralight properties, low-cost manufacturability, and high thrust-to-weight performance. Additionally, to further enhance the bending resistance of the frame of the microrobot, diagonal supports are added between the thrusters (as shown in Fig. 3a); these diagonal supports increase the structural moment of inertia, thereby enhancing the bending resistance of the microrobot.

This study further investigates the performance of differentially metallized emitters, comparing aluminum and tungsten thin-film deposited via magnetron sputtering onto PET substrates. Both metals were uniformly coated to a thickness of 100 ± 10 nm, with precise control of deposition parameters (gas pressure, sputter power, time, etc). Corona discharge experiments were conducted under a voltage of 4.5 kV. Electrical characterization revealed high stability in tungsten-coated emitters, as evidenced by time-dependent current measurements (Fig. 3c). While the aluminum electrode suffered a 58% current reduction after 600 s of continuous corona discharge at 4.5 kV, the tungsten counterpart demonstrated high stability with only 8% current decay under identical conditions. Images of the emitter tips at different times during the corona discharge are shown in Fig. 3d. Comparative analysis revealed evident differences in electrode durability: the aluminum-coated emitter exhibited extensive surface erosion (~90% area affected) after 600 s of operation, while the tungsten counterpart maintained structural integrity with almost imperceptible surface degradation under identical conditions. These results demonstrate that tungsten-coated PET emitters exhibit exceptional erosion resistance during ion wind generation. This performance stems from the intrinsic advantages of the emitter's tungsten coating—including a high melting point (3,422°C), stable work function (4.5 eV), and low sputter yield (<0.1 atoms/ion)—which collectively mitigate ion bombardment damage and maintain consistent electron emission under high electric fields. All these factors will enhance the stability of the flying microrobot, thereby reducing the difficulty of control. Therefore, in this study, all conductive parts of the flying microrobot are sputtered using tungsten. The collective properties of tungsten-coated PET enhance flight stability by minimizing performance degradation, which in turn reduces control complexity. As a result, all conductive components in the microrobot were fabricated using tungsten-coated PET to fully leverage these benefits.

The manufacturing process of the microrobot, as illustrated in Fig. 3e, primarily involves constructing it on a 50- μm -thick PET film substrate. The emitters, collectors, and connectors are precisely cut from the PET film using an ultraviolet (UV) laser cutting system. Subsequently, a 100-nm-thick tungsten layer is sputter-deposited onto the emitters and collectors using a magnetron sputtering system. For assembly, an interlocking structure is employed. Firstly, the connector is inserted vertically into the positioning hole in the collector; the emitter is then placed onto the connector. The precise alignment is ensured by a snap-fit structure. A small amount of ultraviolet-curing adhesive is applied to reinforce the connections. Following manufacturing, a payload flight test was conducted on the microrobot, which verified that its thrust-to-weight ratio reached five and that it could maintain stable flight for over 15 minutes, as documented in Supplementary Movies 3 and 4. Without payload, a continuous tethered flight duration of 1 hour was achieved, as shown in Supplementary Movie 5. In contrast, up-to-date subgram aerial microrobots typically have short lifespans, with hovering durations limited to under 10 seconds⁵³.

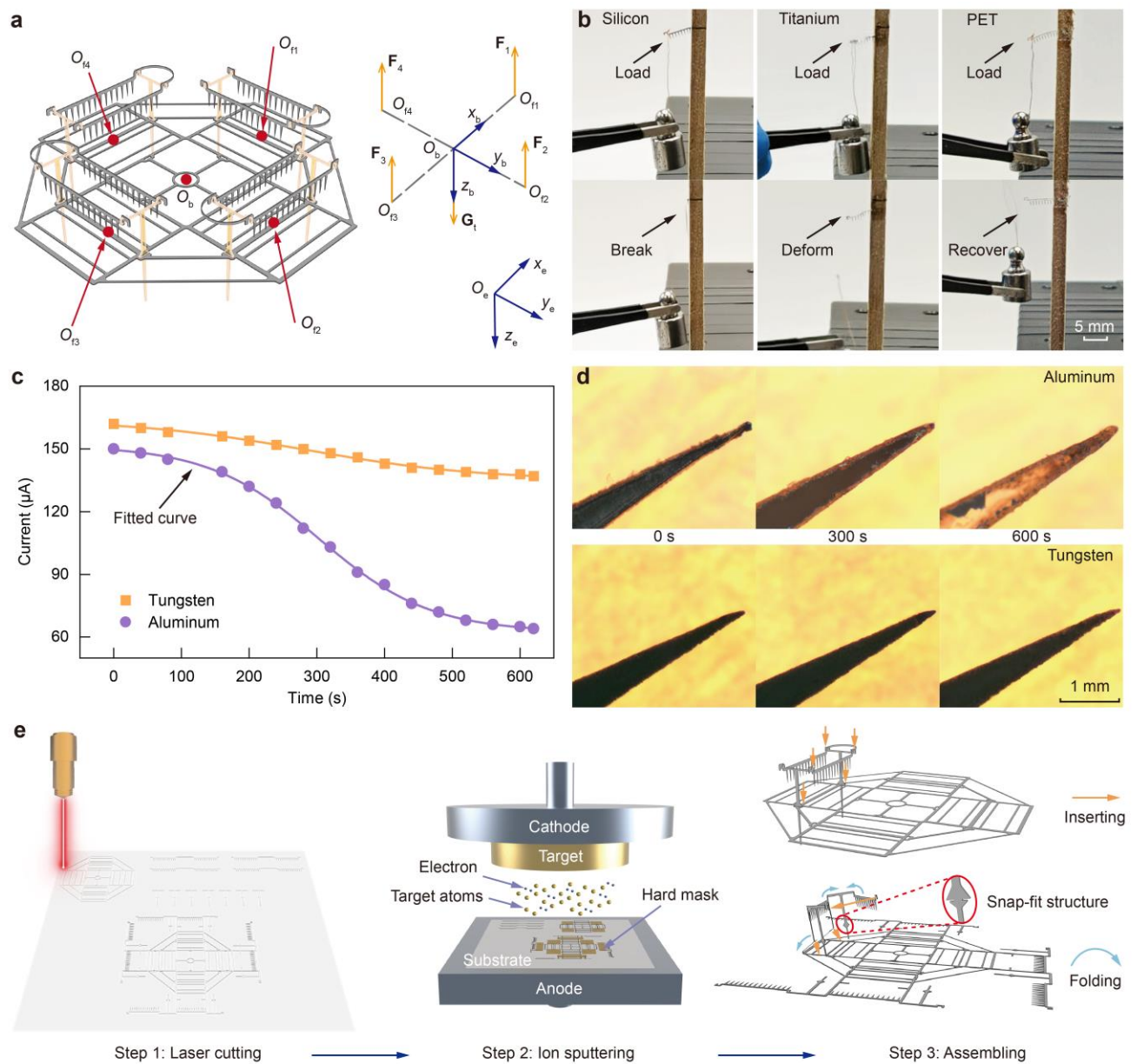


Fig. 3. Performance test of thrusters and manufacturing process. (a) Configuration and force analyses of the designed quadrilateral flying microrobot. (b) Mechanical testing of emitters made of different materials (silicon, titanium, and PET) demonstrates the elasticity and robustness of PET films. (c) Variation of current versus time for emitters coated with aluminum and tungsten. The current of the aluminum-coated emitter decreased by 58%, while it decreased by only 8% for the tungsten-coated emitter. (d) Erosion tests of emitters coated with aluminum and tungsten

during ion wind generation. After a 600-second voltage application, more than 90% of the aluminum coating has spalled, while the tungsten coating remains virtually unchanged, indicating that the tungsten-coated emitter is much more stable than the aluminum-coated one.

(e) Manufacturing process of the proposed flying microrobots, including laser cutting of 50- μm -thick PET films, tungsten ion sputtering, and different assembly methods.

Notably, this work introduces an origami-inspired manufacturing technique that monolithically integrates connectors, emitters, and collectors into a single foldable PET-tungsten composite structure (Fig. 3e), enabling rapid assembly of ion wind-propelled microrobots with fewer discrete components and high precision alignment through self-folding joints. During the magnetron sputtering process, the area of the connector is hard masked with polyimide film, ensuring that only the emitters and collectors are subjected to sputtering. Upon completion of sputtering, the emitters are folded inward, and the connectors are inserted into the small hole of the collectors to complete the assembly. The snap-fit structure guarantees accurate alignment during assembly. This integrated origami-inspired manufacturing technique effectively minimizes structural alignment errors. Both the interlocking and origami-inspired assembly designs improve manufacturing efficiency, enabling complete microrobot assembly in under 10 minutes. The material costs for each microrobot are maintained below \$0.10 USD through the selection of low-cost components. This manufacturing approach satisfies the stringent requirements for the microrobot flight—including mechanical resilience, electrical functionality, and scalable production—while outperforming conventional metal and silicon alternatives in both weight savings and cost efficiency. It thereby facilitates the microrobot's widespread adoption across a broad range of application scenarios.

Flight control design and implementation

Due to its low mass, the ion-propelled flying microrobot is dynamically unstable and highly vulnerable to disturbance during flight. The robot may quickly pitch and roll in the absence of attitude control. The implementation of effective attitude control is a fundamental enabling technology for the flight of ion-propelled microrobots. Owing to its high thrust-to-weight ratio, the proposed microrobot is capable of carrying devices with specific functions. Based on the feedback from an IMU, a closed-loop control algorithm is implemented to stabilize the flight attitude. The angular acceleration of pitch $\ddot{\theta}$ is related to the thrust differences of F_1 and F_3 , while the angular acceleration of roll $\ddot{\phi}$ is related to the thrust differences of F_2 and F_4 . Thus, the angular accelerations of the microrobot are controlled by the voltage difference across the face-to-face arranged thrusters. The control algorithm continuously computes the pitch/roll angular error e and generates the corresponding stabilization thrust vector, which is then converted into precise voltage commands for the ion thrusters. The voltage amplifiers regulate the voltage difference between thrusters 2 and 4 (denoted as U_2), and between thrusters 1 and 3 (denoted as U_3), as dictated by the control signals. The control system achieves active attitude stabilization through real-time voltage modulation of the four ion thrusters, generating precise thrust outputs according to the PID control law.

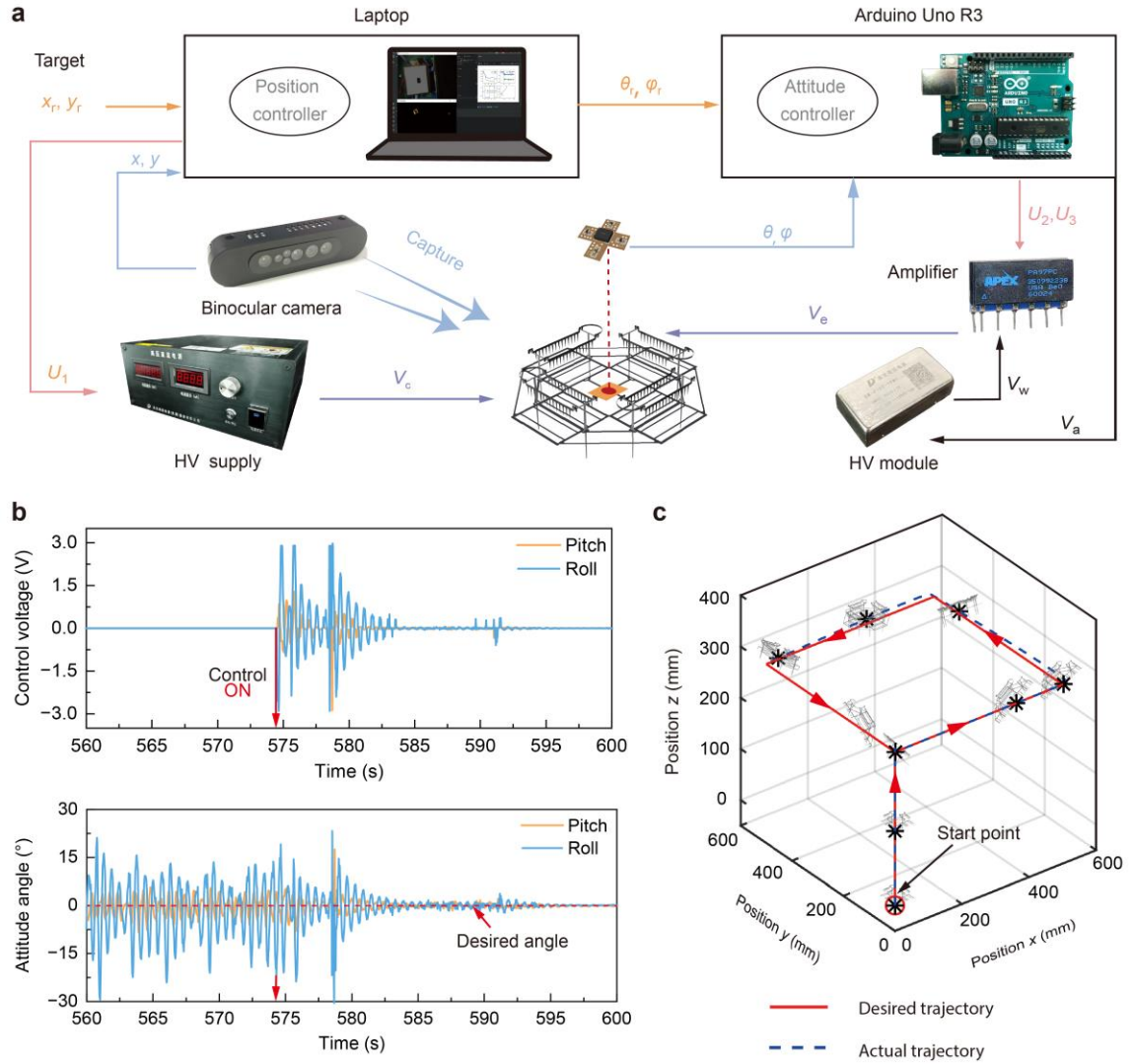


Fig. 4. Control strategy design and verification for the microrobot. (a) Hardware configuration of the control system, including: laptop, microcontroller, binocular camera, and high voltage (HV) supply. (b) Control signals generated by the microcontroller, along with pitch and roll angle measurements from the IMU sensor, show that the control strategy effectively stabilizes the flying microrobot by over 80% improvement in attitude stability under closed-loop control compared to

open-loop operation. (c) Flight simulation with a dual-layer control loop validates ADRC performance: stable vertical takeoff, square-path tracking, and precise return.

The control system and circuit connections are illustrated in Fig. 4a. The negative high-voltage power supply (maximum: -30 kV) provides a reference voltage V_c of -4.5 kV for the collectors, serving as the voltage supply reference for takeoff. The reference voltage is denoted as control signal U_1 , which is associated with the height control strategy and is adjusted based on the feedback current. The emitter voltages are generated by a high-voltage operational amplifier (control range: 0 - 900 V). The amplifier is powered by a high-voltage V_w provided by the HV module with a basic voltage V_a , and controlled by the control signals U_2 and U_3 from a microcontroller (Arduino). The control signal operates within a range of 0 to 4 V. Thus, the operational amplifier operates within a range of 0 to 400 V. Then the system maintains a stable inter-electrode potential difference of 4.5 - 4.9 kV across all four thrust units. Within this optimized voltage range, the thrust exhibits near-linear dependence on applied voltage, as quantified in Fig. 2f. Attitude feedback is provided by an IMU onboard the microrobot, delivering 3-axis angular rate and linear acceleration measurements. After the sensor data are processed, they are fed into a pre-programmed control algorithm in the microcontroller. The control algorithm compares the target attitude angle with the IMU feedback and outputs control signals to the operational amplifier. To achieve absolute coordinate trajectory control without the need for an expensive and complex motion capture system, we propose utilizing a binocular camera to effectively obtain real-time position feedback, enabling the capture and tracking of the flying microrobot's position. The resulting motion trajectory is simultaneously recorded and displayed on a laptop.

Fig. 4b presents the controller's pitch and roll stabilization signals, comparing open-loop (uncontrolled) and closed-loop (controlled) responses. Fig. 4b demonstrates the corresponding

attitude angle measurements from IMU feedback, confirming effective flight stabilization through control voltage adjustment (Supplementary Movie 6). Experimental results demonstrate that without active control, the flying microrobot exhibited unstable lateral flight with attitude deviations exceeding $\pm 30^\circ$ after takeoff. Implementation of closed-loop attitude control successfully stabilized both angular orientation and lateral movement, restricting attitude oscillations to within $\pm 5^\circ$. The settling time to reach and remain within $\pm 5^\circ$ was 6.1 s. The root mean square error (RMSE) of pitch and roll decreased by 83.11% and 89.21%, respectively. The steady-state error of pitch and roll was 0.50° and 1.03° , respectively.

Attitude stabilization permits the use of the small-angle approximation, under which the relationship between the acceleration of the flying robot in the horizontal plane ($\ddot{x}$ and $\ddot{y}$) and its attitude angle can be expressed as:

$$\ddot{x} = -\frac{F_t \theta}{m} \quad (3)$$

$$\ddot{y} = \frac{F_t \varphi}{m} \quad (4)$$

where F_t denotes the total thrust of the four thruster units, and m represents the total mass of the microrobot, including onboard sensors, θ is the pitch angle, and φ is the roll angle. Through this linearized dynamic relationship, the microrobot's flying trajectory can be precisely controlled by regulating its pitch and roll angles. To evaluate the applicability of the established control strategy to the dynamics of an ion-propelled microrobotic platform, we built a reasonable-fidelity MATLAB simulation environment (see Supplementary Fig. 17 for algorithm details). Leveraging this framework, we implemented an active disturbance rejection control (ADRC) scheme to further enhance the anti-disturbance capability of the system. The proposed integrated control strategy enables high-precision trajectory-tracking flight, as demonstrated in Fig. 4c. By dynamically

adjusting attitude angles along preset trajectories, the microrobot achieves stable vertical takeoff, accurately navigates a fixed-altitude square trajectory, and returns to its origin with minimal deviation.

Experimental flight testing and application demonstrations

To evaluate the microrobot's carrying capacity and maneuverability, we conducted two benchmark experiments: payload capacity testing and trajectory tracking assessment. In the payload test, the 36.7 mg robot successfully achieved sustained flight while carrying a 146.8 mg load, demonstrating its high thrust-to-weight ratio of 5:1. The flight process is shown in Fig. 5a and Supplementary Movie 3, which records the flying microrobot hovering in the air with the payload.

In the test for motion along the specified direction, by setting target values for the pitch and roll angles, the microrobot stably adjusts its attitude based on the real-time feedback of the actual attitude angles. The microrobot's tilt directly induces horizontal displacement in the corresponding direction. During the experiment, the microrobot's path data were captured by the binocular camera, and Kalman filtering was employed to mitigate errors arising from the image capture and coordinate calculation processes. The unfiltered signal represents the raw data acquired by the binocular cameras and processed through image processing algorithms. After Kalman filtering, the signal exhibits greater smoothness. Throughout the experiment, real-time adjustments to the target attitude angles enabled the completion of two different laps of the square trajectory, as shown in Fig. 5b. The flying microrobot moves in different directions based on the attitude of the microrobot. The flight time-position curve and the corresponding attitude at each moment are illustrated in Fig. 5b. It can be observed that each variation in the coordinates was a result of attitude adjustments. The experimental process is demonstrated in Supplementary Movie 7, confirming that the ion-propelled flying robot is capable of trajectory tracking. In addition, a flight tracking a triangular

trajectory was completed and demonstrated in Supplementary Movie 8. This process demonstrates that the flying microrobot is capable of performing motion in non-structured orthogonal directions.

After validating both payload capacity and maneuverability, we further explored a critical propulsion-specific challenge regarding whether lightweight sensors can operate reliably despite the intense electric fields inherently generated by EHD thrusters. To assess this compatibility, we integrated two representative sensors—a miniature camera and an FBG sensor, strictly as passive payloads during tethered flight. Rather than demonstrating sensor-driven autonomy, these tests serve to validate the microrobot's electromagnetic compatibility and physical payload capacity in two fundamental scenarios: real-time environmental surveillance and material characterization. For surveillance tasks, the image-sensor-equipped microrobot successfully captured continuous, high-fidelity environmental imagery while maintaining stable flight (Fig. 5c, Supplementary Movie 9). In comparison with the images captured by the imaging sensor prior to takeoff, the recorded images after takeoff exhibited a notable change in perspective.

For material characterization, the FBG-equipped microrobot actively modulated its thrust to enable controlled contact between grating measurement points and target surfaces. The measurement scheme and results are shown in Fig. 5d and Supplementary Movie 10. The microrobot successfully discriminated between material types by analyzing characteristic FBG wavelength shifts at contact points (Fig. 5e). The detailed data are provided in Supplementary Fig. 21. It is evident that for solid materials like foam pads, clay, and chipboard, the wavelength decreases upon contact, with the extent of the wavelength change being more pronounced for objects with higher hardness. When in contact with cobblestone, the FBG sensor undergoes deformation due to the interaction with the smooth surface, leading to an increase in wavelength. For the two liquids, the wavelength first increases and then decreases, with a more significant

decrease observed in the alkaline solution. Quantitative differences in spectral responses enabled clear material differentiation. The characteristic signal differences between different objects exceeded 10 pm, demonstrating the system's high sensitivity to material properties. Highlighting the flying microrobot's versatility, these experimental results demonstrate its potential for environmental surveillance, disaster rescue, and unknown terrain detection, confirming its multifunctionality and practical feasibility for complex tasks.

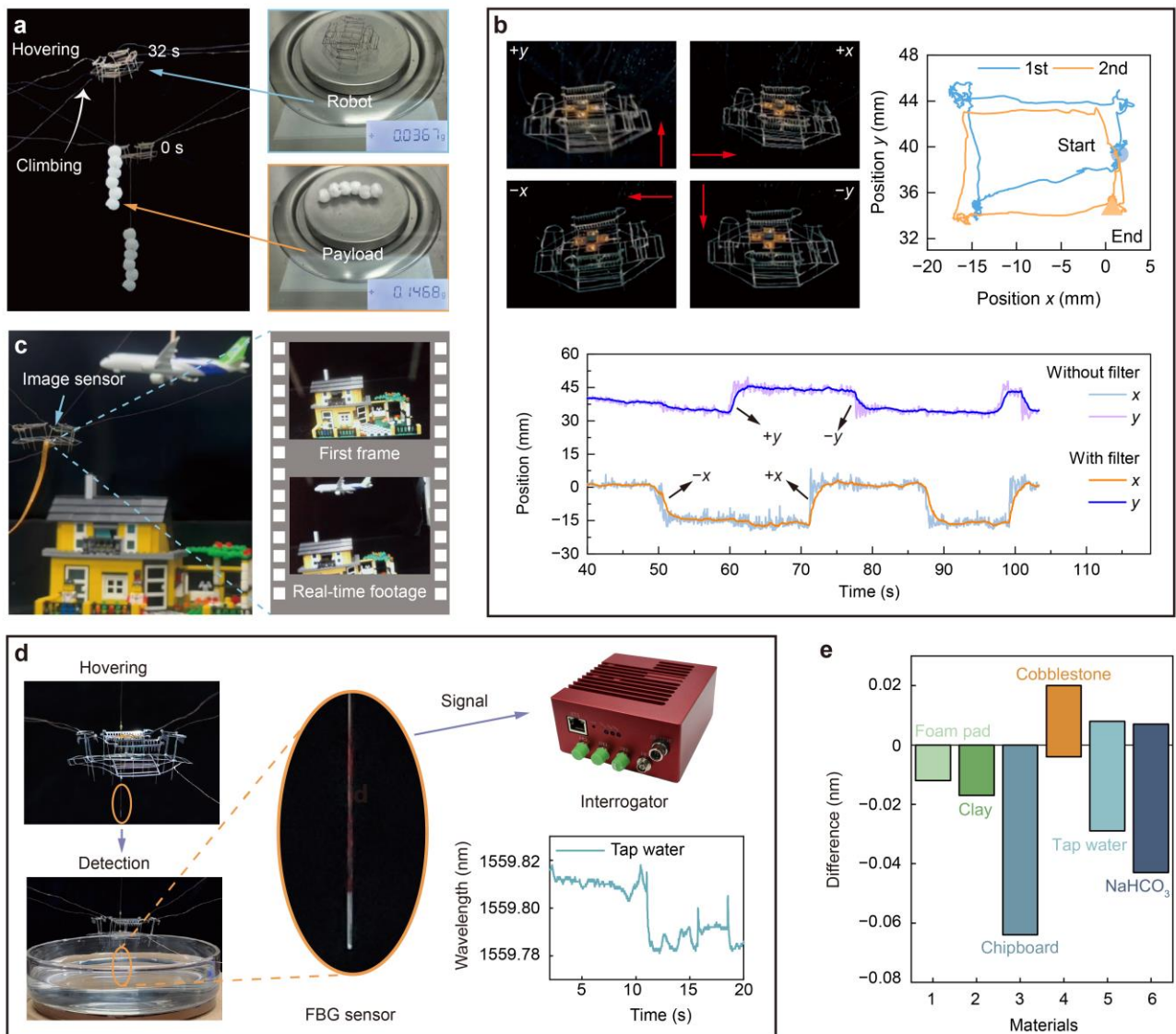


Fig. 5. Controlled flight and application demonstration of the microrobot. (a) The payload capacity test of the 36.7-mg flying microrobot shows that the microrobot can hover in the air with a load of 4 times its own weight. (b) The real-time attitude during turning, as well as the trajectory of the flying microrobot tracking the planar square, and position signals collected by a binocular camera. (c) Images of the flying microrobot equipped with a micro-camera in the center during the robot's flight, and the image transmitted by the image sensor. (d) Demonstration of the flying robot carrying the fiber Bragg grating (FBG) sensor to detect materials: the microrobot equipped with the FBG sensor drops to make contact with the object to be detected; the wavelength variation of the optical fiber is detected by the interrogator, which then converts it into an electrical signal that is transmitted to the laptop for further processing. (e) Difference in wavelength of different detection targets (foam pad, clay, chipboard, cobblestone, tap water, and NaHCO_3 solution) from the FBG sensor. Hard objects (chipboard) corresponded to the greatest wavelength variation, and alkaline substances exhibited larger wavelength changes than tap water.

Discussion

In this work, we present the development of the experimental demonstration of IMU-based closed-loop controlled flight on a centimeter-scale ion-propelled flying microrobot. The proposed ion-propelled flying microrobot generates thrust through ion wind created by corona discharge without mechanical actuators. This design reduces the reliance on complex mechanical components, decreases weight, and opens up possibilities for mass production. The ion-propelled flying microrobot is constructed using a 50- μm -thick tungsten-coated polymer film, innovatively combining the conductivity of the metal with the flexibility of the polymer material. Due to the use of this material, compared to other ion-wind flying robots^{26,36,37} the structure presented in this paper is more complex, but the weight has not increased. As shown in Fig. 1d, conventional

rotorcraft requires electromagnetic motors and propeller structures, which typically results in a larger mass that often exceeds that of ion-propelled microrobots by two orders of magnitude. Even the lightest practical rotorcraft configurations have a mass ($\sim 1.6 \text{ g}^{54}$) that is at least one order of magnitude greater than that of our ion-propelled microrobot. Flapping-wing robots can achieve a lightweight mass on the order of tens of milligrams and have demonstrated a wide range of thrust-to-weight ratios. For instance, the Harvard RoboBee achieved a peak value of 4.1:1⁴⁹. However, ion wind propulsion technology, which operates completely without mechanical moving parts, offers a distinct advantage in reducing structural mass, thereby holding potential in several aspects for the further miniaturization of flying microrobots. To further enhance the flight performance, we also conducted design optimization for the ion-propelled flying microrobot through finite element analysis (FEA). By performing 2D simulations shown in Fig. 2c, the optimal structure ($d_1 = 1.5 \text{ mm}$ and $d_2 = 3.5 \text{ mm}$) was identified for the collector electrodes. The simulation results also indicate that additional emitters have minimal impact on the ion wind velocity. Thus, a single emitter configuration was determined as the optimal design solution. Based on the optimization of the ion thruster, the ion-propelled flying microrobot proposed in this work, weighing 36.7 mg, accomplished the flight carrying the payload of 146.8 mg (Fig. 5a), achieved a high thrust-to-weight ratio of 5. The thrust density of our thruster reaches 538 N/m^3 . This value considerably exceeds the thrust density of 110 N/m^3 reported for the wire-wire ion wind thrusters⁵⁵. While our design shows a lower thrust density compared to recent studies on deeply optimized multi-stage thrusters³² (which can exceed 800 N/m^3), our structural geometry is intentionally optimized for flight dynamics. It should be noted that increasing the distance between the emitter and collector in the structure, or adopting a more vertical multi-stage design, can further enhance the thrust. Since the only change in the structure leading to higher thrust is the lengthening of the connectors,

with no significant increase in the overall structural weight, a higher thrust-to-weight ratio can be achieved. However, increasing the height of the structure is detrimental to flight stability due to the elevated center of gravity and elongation of the structure in the vertical direction, all of which contradict the conception of designing the microrobot with better stability. Furthermore, stability is also influenced by system damping; adding loads like a payload or a dedicated damping structure can effectively enhance it³⁹. Also, the proposed microrobot has longer lifespans compared to other up-to-date sub-gram flying microrobots, since it could maintain stable flight in a tethered state for 1 hour (recorded in Supplementary Movie 5). Compared to other types of robots in terms of weight, thrust-to-weight ratio, and long-duration flight capacity, the robot presented in this paper offers several advantages. Furthermore, 3D simulations of the ion-propelled flying microrobot with different voltage differences (5 kV, 5.5 kV and 6 kV) applied to the emitter and collector electrodes show that the ion wind velocity can be altered by the voltage difference, and the voltage differences applied on different thrusters are independent (Fig. 2a). The optimization of the ion-propelled flying microrobot's structural design through the multiphysics simulation not only enhances the thrust-to-weight ratio but also validates the feasibility of controlling its flight by adjusting the voltage difference.

The high thrust-to-weight ratio and thrust controllability enable the robot to carry an IMU for attitude adjustment through closed-loop feedback with the control strategy shown in Fig. 4a. This capability is critical in addressing the challenges of flying microrobots, particularly regarding flight stability. Traditionally, achieving stable flight in such small flying microrobots has been a challenging task due to their low weight and the inherent instability of ion propulsion at small scales. Using an onboard IMU sensor, the stability of the ion-propelled flying microrobot can be adjusted through real-time feedback⁵⁶. This feedback regulates the emitter's voltage signal, which

in turn alters the EHD force (Fig. 4b). This allows different voltage signals to be applied to the different ion thrusters, enabling multi-degrees-of-freedom motion through the variation in the strength of corona discharge. While previous studies have demonstrated the ability to lift lightweight payloads (e.g., IMU boards) using ion thrusters³⁶, none of those sub-100 mg ion-propelled robots achieved closed-loop controlled flight. This work mounts the IMU sensor on the ion-propelled flying microrobot, achieving stable flight (Fig. 4b and Supplementary Movie 6) and completing controlled path tracking (Fig. 5b and Supplementary Movies 7 and 8). This work verifies the control system by conducting an ion-propelled flying microrobot to fly along rectangular (Fig. 5b and Supplementary Movie 7) and triangular (Supplementary Movie 8) trajectories with closed-loop control, and captures its real-time position using the binocular camera to demonstrate that the miniature ion-propelled microrobot has achieved closed-loop controlled flight with an onboard IMU. Through structural optimization and the design of feedback control systems, this work successfully achieved multi-degrees-of-freedom motion of the ion-propelled flying microrobot. Additionally, an origami technique is employed for the fabrication process of the ion-propelled flying microrobot, shown in Supplementary Fig. 2. Leveraging the origami-inspired design with the cost-effective, high-performance metal-polymer composites, our manufacturing approach enables rapid, simple assembly, overcoming a barrier to swarm deployment.

The microrobot demonstrates a power consumption of approximately 0.45W (4.5 kV, 0.1 mA), yielding a thrust-to-power ratio of around 2.8 mN/W. This efficiency is notably higher than that reported in several recent studies on ionic wind thrusters (e.g., 1.86 mN/W³²). While this power consumption is relatively low for flying microrobots, under current technological constraints, it remains infeasible for these sub-gram microrobots to carry an onboard power supply

sufficient for sustained flight. To bridge this gap, we propose a carrier-deployed system. In this configuration, a larger carrier aircraft provides the necessary power and houses the heavy control electronics, while the tethered microrobots, equipped with onboard sensors, are deployed to perform designated tasks in the confined spaces. This approach enables the flying microrobot to operate efficiently while bypassing the immediate constraints of limited energy capacity. Also, the miniaturization and integration of the control circuit present a significant challenge for flying microrobots. The control system currently relies on relatively heavy electronic components, such as the Apex PA97PC amplifier. Therefore, a key area for further research is to utilize lightweight, compact circuits to detect the environment and generate a controllable high voltage sufficient for regulating ion thrust. This will be essential for improving the performance and feasibility of such robots. The microrobot operates at voltages of 3 to 6 kV, which may raise initial safety concerns. However, the corresponding discharge current remains consistently below 1 mA under normal operating conditions. The voltage and current levels used in this work are already within safe limits for humans⁵⁷. The potential applications of ion-propelled flying microrobots are vast and varied. This study explores several promising avenues for their deployment. In addition to the onboard IMU sensor, functional devices such as an image sensor and an FBG sensor are also mounted. Key areas of interest include payload delivery (Fig. 5a), real-time surveillance (Fig. 5c), and material detection (Fig. 5d) during flight. These applications are valuable in environments where traditional robotic solutions face limitations. When it comes to performing tasks in confined or difficult-to-access spaces, the microrobots demonstrated in this study offer distinct advantages. Their small size and high mobility allow them to accomplish complex tasks such as confined-space surveillance, disaster rescue, and hazardous environment exploration that are otherwise challenging for traditional robots.

Methods

Fabrication of the structure and electrodes

The microrobot's main body was fabricated from 50- μm PET film using a cold-ablation UV laser system (Seaman SM-F10). To minimize heat-affected zones and prevent edge carbonization, we optimized the cutting parameters: 10 W UV power, 10 μm spot size, 10% laser-cutting accuracy, and 100 kHz pulse frequency at 1000 mm/s scan speed. The process required 60 repeat cycles to ensure clean through-cutting while preserving the structural integrity of the thin polymer. Following that, a 100-nm tungsten coating was deposited on the PET-made components via DC magnetron sputtering (Kurt J. Lesker PVD-75) using 140 W power for 10 minutes. Process parameters were optimized to achieve $\pm 5\%$ thickness uniformity across the 50- μm film while maintaining substrate temperature below 80°C to prevent polymer deformation. The main body and components of the ion-propelled flying microrobot were assembled and structurally bonded at the joints using a localized UV-curable adhesive (EMI 10483). This bonding ensures the mechanical integrity essential for flight maneuvers. The complete fabrication sequence is detailed in Supplementary information.

Performance characterization of the ion wind thrusters

The experimental characterization of the ion wind thruster's velocity and thrust is detailed in Supplementary information (Supplementary Figs. 7 and 13). Ion wind velocity was measured using a calibrated thermal anemometer (Testo 405i; 0.01 m/s accuracy) mounted on a manual translation stage and positioned 20 mm downstream of the thruster's collector plane. A voltage difference was then applied between the collectors and emitters, causing the ion wind produced by corona discharge to be recorded by the anemometer.

Thrust measurements were performed using a precision balance (XIULAB FA2204; 0.1 mg accuracy) with the ion thruster mounted above a grounded electrostatic shield that fully enclosed the balance. This shielding configuration effectively blocked ionic interference while allowing accurate force measurement of the thruster output. Additionally, a sufficient distance should be maintained between the ion wind thruster and the electrostatic shield to further prevent the ion wind from impacting the precision balance, thereby minimizing measurement errors. The net lifting force generated by the thruster was transduced into a measurable weight loss of the support structure, enabling direct thrust quantification. Then the weight loss was finally converted into thrust.

Ion wind flow was visualized in a custom low-speed wind tunnel ($Re < 100$), revealing the three-dimensional flow structures generated by the microrobot during operation. A smoke generator was placed at the inlet of the wind tunnel, and a smoke extractor was placed at the outlet. Additionally, sponges were installed at both the inlet and outlet to ensure a laminar and uniform flow of the working fluid. During the experiment, the flying microrobot was suspended at the center of the wind tunnel using 30- μ m enameled wire. Uniform smoke flow patterns were achieved by precisely balancing the smoke concentration at the inlet (10-15% by volume) with the extraction velocity at the outlet, enabling clear visualization of the ion wind interaction field surrounding the microrobot.

Multiphysics simulation to design emitter and collector

Multiphysics simulation optimizes the emitter and collector structures of an ion-propelled flying microrobot to improve key performance metrics such as thrust-to-weight ratio. First, a multiphysics coupling approach is used to couple the electrostatic field, laminar flow field, and rarefied transport field. The rarefied transport field characterizes the motion of ions, while the

laminar flow field characterizes the ion wind. Assuming the fluid is an incompressible Newtonian fluid, the flow can be characterized by the Navier-Stokes equation²⁰:

$$\rho(\mathbf{u} \cdot \nabla)\mathbf{u} = -\nabla \mathbf{p} + \nu \nabla^2 \mathbf{u} + \mathbf{F}_t \quad (5)$$

where ρ is the fluid density, $\mathbf{u}$ represents the fluid velocity vector field, $\mathbf{p}$ is the pressure, ν is the kinematic viscosity, and $\mathbf{F}_t$ is the Coulomb force related to the charge density ρ_e and the electric field intensity $\mathbf{E}$.

The relationship between the electric field intensity and the spatial charge density can be described by Gauss's law:

$$\nabla \mathbf{E} = \frac{\rho_e}{\varepsilon} \quad (6)$$

where ε is the dielectric permittivity. Then, considering the relationship between the electric field intensity and the electric potential, Poisson's equation is derived:

$$\nabla^2 V = -\frac{\rho_e}{\varepsilon} \quad (7)$$

where V is the electric potential. The current density $\mathbf{J}$ of the electric current generated by charge drifting is expressed as:

$$\mathbf{J} = \mu \mathbf{E} \rho_e + \mathbf{u} \rho_e - D \nabla \rho_e \quad (8)$$

where μ is the ion mobility, and D is the ion diffusion coefficient.

The above equations were numerically solved within the Laminar Flow, Transport of Diluted Species, and Electrostatics modules of the Comsol Multiphysics software. The boundary conditions for the computational domain were set by $\mathbf{p} = 0$. The specific simulation parameters are detailed in Supplementary Table 5. Next, steady-state analysis is used to determine the impact of different emitter and collector structures on the ion wind velocity, identifying the style B collector layout as optimal. A comparative analysis of different emitter quantities revealed that multiple

emitters do not effectively increase the ion wind velocity. Additionally, three-dimensional steady-state and transient simulations were conducted, and a parametric scan of the ion wind velocity of the entire ion-propelled flying microrobot was performed. Different voltages were applied to various emitter structures. This approach allows for verification of the impact of different voltages on ion wind speed from an overall perspective, providing theoretical validation and planning for control experiments.

Closed-loop microrobot flight control

First, a negative bias voltage was applied to the collector to reach the corona discharge voltage level using a high-voltage power supply (DONGWEN DW-P503-4ACD2). The four thrusters were connected to four high-voltage operational amplifiers (APEX PA97PC) with a gain of 100, controlled by a microcontroller (Arduino Uno R3). The operational amplifiers were powered by a high-voltage module (DONGWEN DW-P102-1FN1), which required both a 24 V DC power supply and a 0-5 V Arduino control signal for precise voltage regulation. Effective control of the microrobot was achieved by amplifying a control signal of up to 4 V from the Arduino by a factor of 100 using high-voltage operational amplifiers, with the high-voltage module DW-P102-1FN1 providing a 400 V working voltage to the amplifiers. During flight, positive voltages were applied to the emitters through four fine (30- μm) enameled copper wires to minimize aerodynamic interference, while a single shared wire delivered negative voltages to all collectors. The wires are used exclusively for power supply and signal transmission. A shelf was mounted on the robot, with a flexible PCB housing a six-axis IMU (STMicroelectronics LSM6DS3TR) positioned at its center. Four 30- μm diameter enameled copper wires were employed to transmit data from the sensor to the Arduino, facilitating real-time monitoring of the microrobot's flight attitude with an IMU sampling rate of 100 Hz. The Kalman filter was implemented in the Arduino program to estimate

the system's state, thereby mitigating the effects of drift and providing more accurate readings over extended periods of flight. The onboard control-loop frequency for attitude regulation was also set to 100 Hz. A binocular stereo camera (WHEELTEC; <200 USD) was mounted 20 cm above the experimental arena to track the microrobot's 3D absolute position with 0.1-mm spatial resolution at 60 Hz. The position control loop was executed at 10 Hz. The circuit is shown in Supplementary Fig. 20. In the experiments of surveillance and material characterization, the image sensor (Yichuang Electronics EZ-EN35L-S), FBG sensor (Tongwei), and interrogator (Tongwei, TV1600-mini) are required.

Data availability:

All data supporting the findings and conclusions of this study are provided in the main text and Supplementary information. The raw imaging data is available from Zenodo under <https://doi.org/10.5281/zenodo.20994031>.

Code availability:

Computer code is available from Zenodo under <https://doi.org/10.5281/zenodo.21072662>.

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Author contributions:

Q.T., Y.G., and W.L. led the development of the concepts and designed the experiments. Q.T. and Y.G. interpreted the results and wrote the paper. W.L., Y.C., B.H., and X.C. led the development of the methodology and edited the paper. W.L. acquired the funding and supervised the research. X.W., F.L., F.X., T.B., and K.T. conducted the experiments. X.W. fabricated the robots. F.L., F.X., and T.B. fabricated the circuits and developed codes for the control system. Z.C. performed the simulations. Y.Z. provided the FBG and the interrogator, and conducted the detection experiments.

Competing interests:

The authors declare no competing interests.